National University of Computer and Emerging Sciences Islamabad
Operating System Lab Spring 2025

Lab Task 1
Basic Linux Commands, File Permissions, Conditional Statements

---

Practice Resource

To help you practice shell scripting, refer to the following link with practical examples:
Shell Script Examples - GeeksforGeeks

This resource provides hands-on examples of shell scripts that will enhance your understanding of the topics covered in this lab. Ensure you explore and try out the examples before attempting the tasks**.**

Basic Linux Commands: Lab Tasks

Task 1: File Content Manipulation

1. Create a new file named `system_info.txt`.
2. Append the text "Linux is versatile!" to the file.
3. Display only the first line of the file in the terminal.

Task 2: Directory Navigation and Copying

1. Create a directory named `projects_backup`.
2. Copy the `system_info.txt` file into the `projects_backup` directory.
3. Navigate to the `projects_backup` directory and confirm the file exists by listing all files.

Task 3: File Renaming and Cleanup

1. Create three files: `temp1.log`, `temp2.log`, and `temp3.log`.
2. Rename `temp2.log` to `backup.log`.
3. Delete `temp1.log` and verify only `temp3.log` and `backup.log` remain.

Linux File Permissions: Lab Tasks

Task 1: Changing Permissions

1. Create a file named `experiment.txt`.
2. Modify its permissions to:
    - Allow the owner full access (read, write, execute).
    - Allow the group read-only access.
3. Verify the changes using `ls -l`.

Task 2: Adjusting Permissions Dynamically

1. Create a file named `user_data.csv`.
2. Grant write permissions to the group.
3. Remove all permissions for others (no read, write, or execute).
4. Verify each change using `ls -l`.

Loop Task: While Loop

Question:
Write a shell script to:

1. Ask the user to enter a number.
2. Use a **while loop** to print all even numbers starting from 1 up to the entered number.
3. Display "Task completed!" once the loop finishes.

Conditional Task: Case Statement

Question:
Write a shell script to:

1. Prompt the user to input a day of the week (e.g., "Monday", "Tuesday").
2. Use a **case statement** to display:

    - "Start of the week!" for Monday.
    - "Midweek hustle!" for Tuesday to Thursday.
    - "Weekend vibes!" for Friday to Sunday.
3. Handle invalid inputs with "Invalid day entered. Please try again."

Instructions for Students

1. Complete all tasks in sequence.
2. Write the required scripts for conditional and loop tasks in separate files.
3. Test each task in the terminal and document the results.
4. Submit your completed scripts and terminal outputs for review.

THE END